Sec: SR.IIT_*COSC(MODEL-A) | SPCL GTM-13 | Date: 25-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**A**	2	**B**	3	**B**	4	**C**	5	**D**
6	**C**	7	**C**	8	**A**	9	**C**	10	**D**
11	**B**	12	**C**	13	**B**	14	**C**	15	**A**
16	**D**	17	**A**	18	**B**	19	**C**	20	**B**
21	**6**	22	**2**	23	**324**	24	**75**	25	**3**

CHEMISTRY

26	**A**	27	**D**	28	**A**	29	**D**	30	**C**
31	**D**	32	**D**	33	**B**	34	**A**	35	**A**
36	**B**	37	**C**	38	**A**	39	**D**	40	**B**
41	**A**	42	**D**	43	**C**	44	**B**	45	**B**
46	**25**	47	**2**	48	**2**	49	**57**	50	**10**

MATHEMATICS

51	**C**	52	**A**	53	**C**	54	**B**	55	**C**
56	**C**	57	**C**	58	**A**	59	**B**	60	**C**
61	**C**	62	**C**	63	**C**	64	**A**	65	**A**
66	**B**	67	**B**	68	**B**	69	**B**	70	**D**
71	**998**	72	**17**	73	**35**	74	**2**	75	**1**

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SOLUTIONS

PHYSICS

1. Given $\alpha = \frac{Fv^2}{\beta^2}\left[\log_e\left(\frac{2\pi\beta}{v^2}\right)\right]$

$\downarrow$

Dimensionless

Dimensions of $\frac{2\pi\beta}{v^2} = 1$

$\therefore$ Dimensions of β = dimensions of v^2

$= \left[LT^{-1}\right]^2 = \left[L^2T^{-2}\right]$

Dimensions of $\alpha = \frac{Fv^2}{\beta^2}$

$= \frac{\left[MLT^{-2}\right]\left[LT^{-1}\right]^2}{\left[L^2 - T^{-2}\right]^2} = \left[M^1L^{-1}T^0\right]$

2. Pitch=1 mm

Number of division on circular scale=200

$$LC = \frac{\text{pitch}}{\text{number of divisions on circular scale}}$$

$= \frac{1\text{mm}}{200} = 0.005\text{mm} = 0.0005\text{cm}$

Diameter of the wire= (Main scale reading + Circular scale reading $\times$LC) -zero error

$= 6\text{mm} + 45 \times 0.005 - (-0.05)$

$= 6\text{mm} + 0.225\text{mm} + 0.05\text{mm}$

$= 6.275\,\text{mm}$

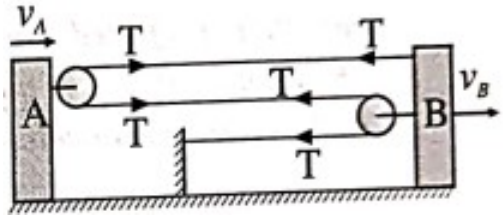
3. $Y_1 = A + B$, $Y_2 = \overline{A.B}$

$Y = (A+B).\left(\overline{A.B}\right)$

$= A.\overline{A} + A.\overline{B} + B.\overline{A} + B.\overline{B}$

$= 0 + A.\overline{B} + B.\overline{A} + 0 = XOR$

4. Given, $v_A - v_B = v$



or $v_A = (v_B + v)$

Using, $\sum Tv\cos\theta = 0$

or $(2T)v_A\cos 0^0 + (3T)v_B\cos 180^0 = 0$

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or $2Tv_A - 3Tv_B = 0$

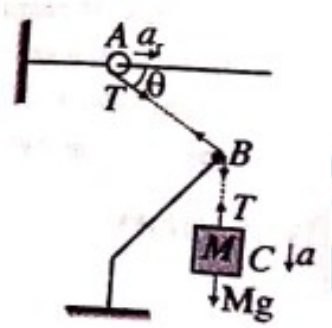
or $v_A = \frac{3}{2}v_B = \frac{3}{2}(v_A - v)$

or $v_A = 3v$

Acceleration of block A, $a_A = \frac{v_A - 0}{t} = \frac{3v}{t}$

5. If a is the acceleration of block and a_r is the acceleration of the ring, them

$Ta_r \cos\theta + Ta\cos 180^0 = 0$



or $a = a_r \cos\theta$

or $a_r = a/\cos\theta$ .....(i)

Now using Newton`s second law, we have

$Mg - T = Ma$ .....(ii)

and $T\cos\theta = ma_r$....(iii)

On solving above equations, we get

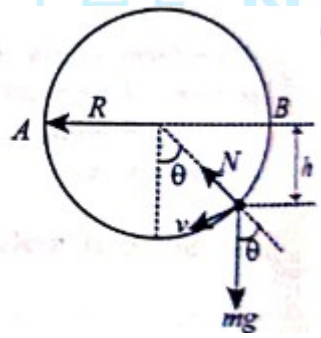
$a = \frac{Mg\cos^2\theta}{m + M\cos^2\theta}$

and $a_r = \frac{a}{\cos\theta} = \frac{Mg\cos\theta}{m + M\cos^2\theta}$

Putting $M = 2m$ and $\theta = 30^0, g = 10m/s^2; a = 6m/s^2$, $a_r = 4\sqrt{3}m/s^2$

6. $v = \sqrt{2gh}$

$= \sqrt{2gR\cos\theta}$.



Now, $N - mg\cos\theta = \frac{mv^2}{R}$

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or $N = mg\cos\theta + \frac{mv^2}{R}$ ....(i)

Given $K = \frac{1}{2}mv^2 = \frac{1}{2}m \times 2gR\cos\theta$

From above equations $= mgR\cos\theta$ ….(ii)

$$N = \frac{K}{R} + \frac{2K}{R} = \frac{3K}{R}$$

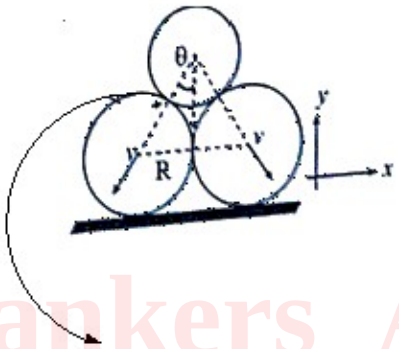
7. At the instant of collision, the situation is shown in figure from geometry,

$$\sin\theta = \frac{R}{3R/2} = \frac{2}{3}$$

Using conservation of momentum in vertical direction,

$mv_0 = 2m \times v\cos\theta$

or $v = \left(\frac{v_0}{2\cos\theta}\right)$



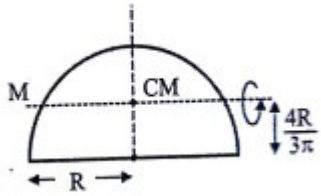
$$R + \frac{R}{2} = \frac{3R}{2}$$

Now horizontal velocity,

$v_x = v\sin\theta$

$$= \frac{v_0}{2\cos\theta} \times \sin\theta = \frac{v_0}{2}\tan\theta = \frac{v_0}{\sqrt{5}}$$

8. Using $I_D = I_{CM} + md^2$



or $I_{CM} = I_D - md^2$

$$= \frac{mR^2}{4} - m\left(\frac{4R}{3\pi}\right)^2 = mR^2\left[\frac{1}{4} - \frac{16}{9\pi^2}\right]$$

9. Using conservation of angular momentum and ME, we have

$mu(R+h) = mvR\sin\theta$ …….(i)

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and $\frac{1}{2}mu^2-\frac{GMm}{R+h}=\frac{1}{2}mv^2-\frac{GMm}{R}$ .....(ii)

Substituting h=R and $u=\sqrt{\frac{GM}{7R}}$, and simplifying, we get



$\sin\theta=\frac{1}{\sqrt{2}}$

or $\theta=45^0$

10. Using, $\frac{\Delta V}{V}=\frac{\Delta L}{L}(1-2\sigma)$

Here $\Delta L=\frac{FL}{AY}$,

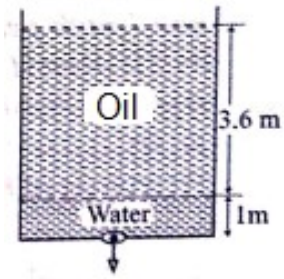
$\therefore \quad \frac{\Delta V}{V}=\frac{F}{AY}(1-2\sigma)$, and $\Delta V=\frac{F}{AY}(1-2\sigma)AL$

$\Delta V=(1-2\sigma)\frac{FL}{Y}$

11. Height of water which exerts the same pressure on interface, whatever oil exerts, let it is h.

$\therefore \quad h\rho_w g=4\times\rho_{oil}\times g$

or $h=\frac{4\times900}{1000}=3.6m$



Effective height of water over the hole

H=1+3.6=4.6 m

$v_c=\sqrt{2gh}=\sqrt{2\times9.8\times4.6}$

=9.5 m/s

12. Using, $K_e=\frac{K_1A_1+K_2A_2}{A_1+A_2}$

Here $A_1=\pi R^2$ and $A_2=\pi(4R^2-R^2)=3\pi R^2$

$\therefore \quad K_e=\frac{K\times\pi R^2+(2K)\times3\pi R^2}{\pi R^2+3\pi R^2}=\frac{7}{4}K.$

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13. Given, $P = aV^b$

or $PV^{-b} = a$

Compare with $PV^r = \text{Constant}$, we have

$r = -b$

We know that, $C = C_V - \dfrac{R}{r-1}$

Here, $C = 0, C_V = \dfrac{R}{\gamma - 1}$

$\therefore \quad 0 = \dfrac{R}{\gamma - 1} - \dfrac{R}{-b-1}$

or $b = -\gamma$

14. The effective force constant

$$k_e = \underbrace{(k+k)}_{\text{vertical springs}} + \underbrace{2(2k)\cos^2 45^0}_{\text{Inclined springs}}$$

$$= 2k + 2 \times 2k \times \frac{1}{2} = 4k.$$

Time period, $T = 2\pi\sqrt{\dfrac{M}{4k}} = \pi\sqrt{\dfrac{M}{k}}$

15. Suppose L_0 and L_C are the lengths of open pipes respectively.

Frequency of first overtone of open organ pipe,

$$f_0 = \frac{2v}{2L_0} = \frac{v}{L_0}$$

Frequency of first overtone of close organ pipe

$$f_C = \frac{3v}{4L_C}$$

Given $f_0 - f_C = 2.2 \text{ Hz}$

$$\therefore \quad \frac{v}{L_0} - \frac{3v}{4L_C} = 2.2$$

As $\dfrac{v}{4L_c} = 110 \text{ Hz}$ and $v = 330 \text{ m/s}$

$$\therefore \quad \frac{330}{L_0} - 3 \times 110 = 2.2$$

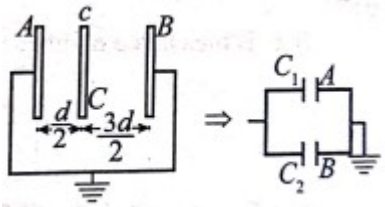
or $L_0 = 0.99\text{m}, L_C = 0.75\text{m}$

16. If q_1 is the charge on plate A

q_2 on B, then $q_1 + q_2 = q \quad \ldots\ldots(i)$

and $\dfrac{q_1}{C_1} = \dfrac{q_2}{C_2} \quad \left[\overline{V} = V_2\right]$

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or $\dfrac{q_1}{\left(\dfrac{\epsilon_0 A}{d/2}\right)} = \dfrac{q_2}{\left(\dfrac{\epsilon_0 A}{3d/2}\right)} \ldots\ldots (ii)$

On solving above equation, we got $q_1 = \dfrac{3q}{4}$ and $q_2 = \dfrac{q}{4}$.

17. $V_A - V_B = \left(\dfrac{\varepsilon}{4+6}\right)\times 4 = \dfrac{4\varepsilon}{10}$ (gn series I = Consta $V\alpha R$)

$V_A - V_D = \dfrac{\varepsilon}{6+4}\times 6 = \dfrac{6\varepsilon}{10}$

$V_B - V_D = (V_A - V_D) - (V_A - V_B) = \dfrac{6\varepsilon}{10} - \dfrac{4\varepsilon}{10} = \dfrac{\varepsilon}{5}$

18. Momentum, $P = \int F dt$

$= \int (Bid)\, dt$

$= Bd\int i dt$

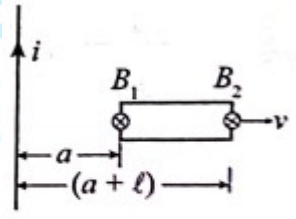
$= BdQ_0$

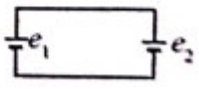
19. When, x=a, the position of the loop is shown in figure. The magnetic field at left arm of the loop

$$B_1 = \frac{\mu_0 i}{2\pi a}$$

Similarly, magnetic field at right arm of the loop

$$B_2 = \frac{\mu_0}{2\pi}\frac{i}{(a+\ell)}$$





Now induced emf across left and right arms are

$e_1 = B_1 vb$ and $e_2 = B_2 vb$

$e_{net} = e_1 - e_2$

$= vb[B_1 - B_2]$

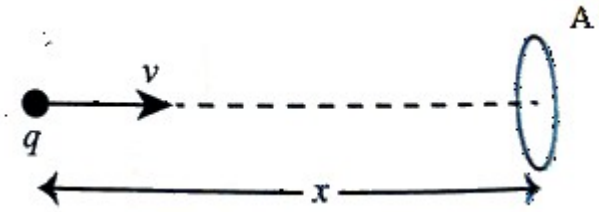
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$$= vb\left[\frac{\mu_0}{2\pi}\frac{i}{a} - \frac{\mu_0}{2\pi}\frac{i}{(a+\ell)}\right]$$

$$= \frac{\mu_0 i(b\ell v)}{2\pi a(a+\ell)}$$

20. Displacement current

$$i_d = \epsilon_0 \frac{d\phi_e}{dt}$$



$$= \epsilon_0 \frac{d(EA)}{dt}$$

$$= \epsilon_0 A\left(\frac{dE}{dt}\right)$$

$$= \epsilon_0 A\frac{d}{dt}\left(\frac{1}{4\pi\epsilon_0}\frac{q}{x^2}\right)$$

$$= \frac{qAv}{2\pi x^3} \qquad \left(\frac{dx}{dt} = v\right)$$

21. The RI of quartz, $\mu_1 = \frac{3\times10^8}{1.5\times10^8} = 2$

RI of glycerin, $\mu_2 = \frac{3\times10^8}{\left(\frac{9}{4}\right)\times10^8} = \frac{4}{3}$

Shift produced, $S = t\left(1 - \frac{1}{\left(\frac{\mu_1}{\mu_2}\right)}\right)$

$$= t\left[-\frac{\mu_2}{\mu_1}\right]$$

$$= 18\left[1 - \frac{\left(\frac{4}{3}\right)}{2}\right] = 6 \text{ cm.}$$

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22. Using, $I = 4I_0 \cos^2 \frac{\phi}{2}$

or $\frac{3(4I_0)}{4} = 4I_0 \cos^2 \frac{\phi}{2}$

$\therefore \cos\frac{\phi}{2} = \frac{\sqrt{3}}{2} \Rightarrow \frac{\phi}{2} = \frac{\pi}{6}$ or $\phi = \frac{\pi}{3}$

Now using $\Delta x = \frac{\lambda}{2\pi}\phi$

or $\frac{dy_1}{D} = \frac{\lambda}{2\pi} \times \frac{\pi}{3}$

$\therefore y = \frac{D\lambda}{6d}$

The minimum distance between two maxima

$= 2y_1 = \frac{2D\lambda}{6d} = \frac{1\times 600\times 10^{-9}}{3\times 10^{-3}} = 0.2$ mm

23. $0 = \vec{P}_{photon} + \vec{P}_H$

or $\vec{P}_H = -\vec{P}_{photon}$

or $P_H = P_{photon}$

or $mv = \frac{h}{\lambda}$

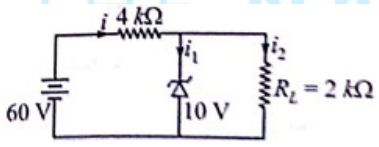
$v = \frac{h}{m\lambda} = \frac{6.63\times 10^{-34}}{1.67\times 10^{-27}\times 122\times 10^{-9}} \simeq 3.25$ m/s

$= 325\times 10^{-2}$ m/s

24. The current in load resistor R_L

$i_2 = \frac{V}{R_L} = \frac{10}{2\times 10^3} = 5$ mA

The current $i = \frac{V - V_z}{R}$



$= \frac{60-10}{4\times 10^3} = 12.5$mA

The current $i_1 = i - i_2 = 12.5 - 5 = 7.5$ mA.

$= 75\times 10^{-4}$ A

25. Relative separation $= V_{BA} \times t$

$8\times 3 = (10-2)\times t$

$t = 3$s

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CHEMISTRY

26. $K_b = \dfrac{RT_b^2}{1000L_V}$ $L_v = \dfrac{\Delta H_V}{M}$

$= \dfrac{1.987 \times 373 \times 373}{1000 \times 540}$ M = MW of solvent

$= 0.513\,K - kg - mol^{-1}$

27. $[R] = [R]_0\ e^{-kt}$

(OR)

$\ell n[R] = \ell n[R]_0 - kt$ $[R]_0$ = Initial Concentration

y = c +mx

28. Rhamnose is 6-deoxy glucose
29. Conceptual
30. Conceptual
31. $A_2 \rightarrow 2A$

1 0

$1-0.2$ 0.4 Total no of moles=1.2

=0.8

$P_{A_2} = \dfrac{0.8}{1.2} \times 1 = \dfrac{2}{3}$

$P_A = \dfrac{0.4}{1.2} \times 1 = \dfrac{1}{3}$

$K_P = \dfrac{P_A^2}{P_{A_2}} = \dfrac{1}{6}$

$\Delta G^0 = -2.303\ RT \log K_P$

$= -2.303 \times 8.314 \times 320 \log \dfrac{1}{6} = 4767\ J$

32. $Ag + Ag^+(0.1\,M) \rightarrow Ag^+(C_2) + Ag$

$E = E^0 - \dfrac{0.06}{n} \log \dfrac{C_2}{C_1}$

$0.162 = 0 + \dfrac{0.06}{1} \log \dfrac{C_1}{C_2}$

$2.7 = \log \dfrac{0.1}{C_2}$

$2.7 = \log 0.1 - \log C_2$

$[Ag^+] = C_2 = 2 \times 10^{-4}$

$Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{-2}$

2S S

$2S = 2 \times 10^{-4}$

$S = 10^{-4}$

$\therefore Ksp = 4S^3$

$= 4(10^{-4})^3 = 4 \times 10^{-12}$

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33. $\lambda = \frac{h}{\sqrt{2mKE}}$

$\lambda \alpha \frac{1}{\sqrt{KE}}$

$\lambda \alpha \frac{1}{\sqrt{T}} \quad (\because KE \alpha T)$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}}$

$\frac{\lambda}{\lambda_2} = \sqrt{\frac{1200}{300}} = 2$

$\lambda_2 = \frac{\lambda}{2}$

34. $\Delta H_{ionisation \text{ of WA}} = \Delta H_{neutralisation \text{ of WA \& SB}} - \Delta H_{neutralisation \text{ of SA \& SB}}$

$= -25.4 - 2(-13.7)$

$= -25.4 + 27.4$

$= 2$ K.Cal

35. O_2 = 16 electrons – Paramagnetic

N_2 = 14 electrons – Dia

CN^- = 14 electrons – Dia

B_2 = 10 electrons – Paramagnetic

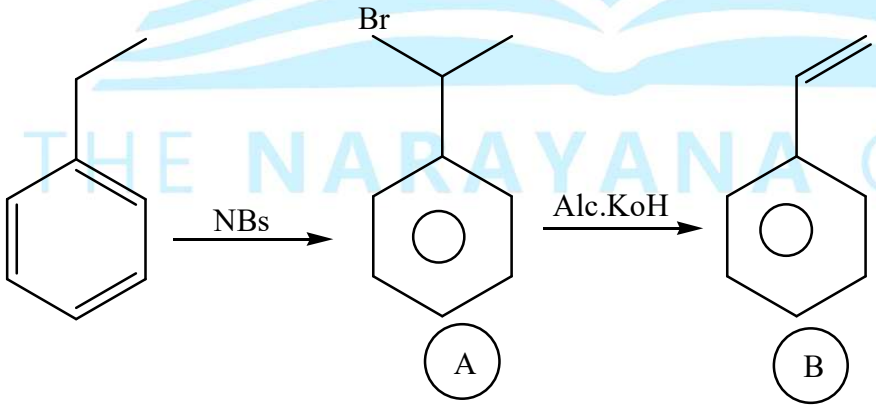
F_2 = 18 electrons – Dia

O_2^+ = 15 electrons – Paramagnetic

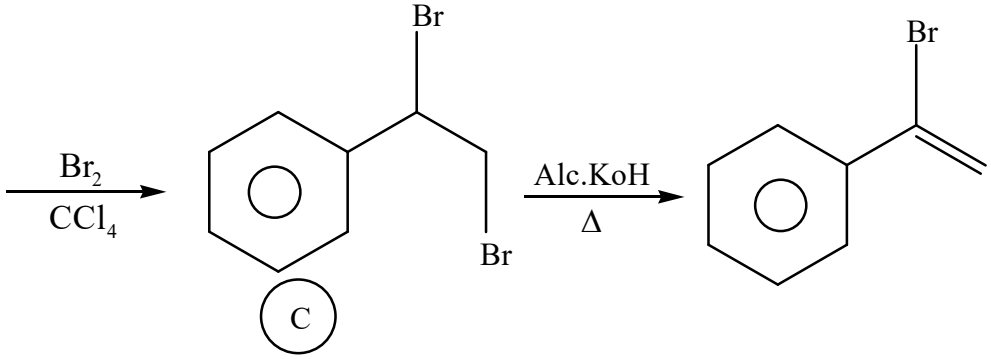
$CN^-, N_2 = 14$ electrons

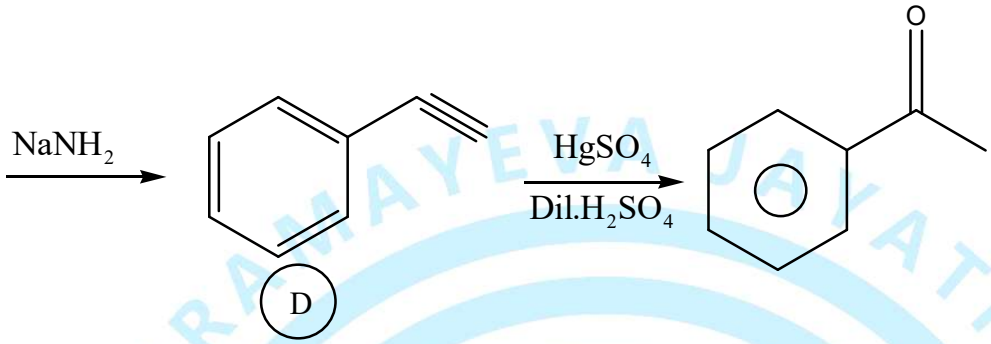
Bond Order $= \frac{Nb - Na}{2} = \frac{10 - 4}{2} = 3$

36.



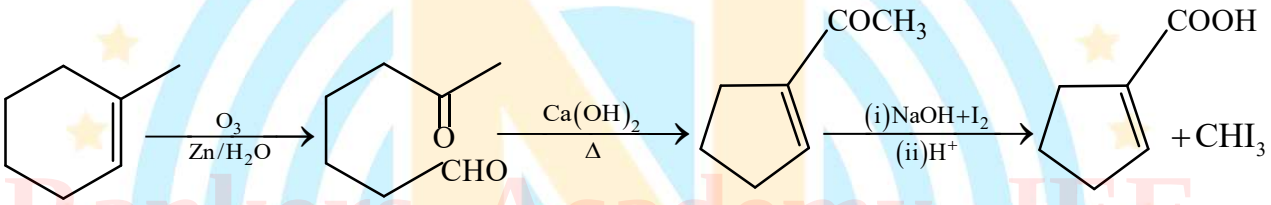
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37. Conceptual

38.



39. Conceptual

40. The element is nitrogen it forms most acidic oxide, N_2O_5

41. Conceptual

42. $\%N = \dfrac{1.4 \times V_a \times Na}{\text{wt of } 0.C} \Rightarrow 7 \times x = \dfrac{1.4 \times 0.2 \times 10^{-4} \times 10^{-3} \times 2}{0.002}$

$\Rightarrow 7 \times x = \dfrac{1.4 \times 0.2 \times 10^{-1} \times 2}{0.002} \Rightarrow 7 \times x = \dfrac{14 \times 2}{100 \times 2} \times 10^{-1} \times 10^{3} \times 2$

mol $\rightarrow$ millimolar Na

$\Rightarrow 7 \times x = 28_4 \Rightarrow x = 4$

43. $M_{wt} = 2 \times V.D$

$= 2 \times 117 = 234$ (polyester compound)

$$\text{Number of OH groups} = \frac{\text{Polyester molecular weight} - \text{Molecular weight of polyalcolic comound}}{42}$$

$$= \frac{234 - 150}{42} = \frac{84}{42} = 2$$

44. $Cr^{+3}, Ni^{+2}, Fe^{+2} \rightarrow$ Green

$Fe^{2+} \xrightarrow{H_2S} FeS \xrightarrow{HCl} FeCl_2$

45. Conceptual

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46. $PCl_5 \rightarrow PCl_3 + Cl_2$

1	0	0	P	300K	1 mole
1-x	x	x	2P	480K	(1+x mole)

$PV = nRT$

$P \alpha nT$

$$\frac{P_1}{P_2} = \frac{n_1}{n_2}\frac{T_1}{T_2}$$

$$\frac{P}{2P} = \frac{1}{1+x}\frac{300}{480}$$

$$\frac{1}{2} = \frac{1}{(1+x)}.\frac{30}{48}$$

$$(1+x) = \frac{60}{48} = 1.25 \qquad x = 0.25$$

% DOD=25

47. $C_nH_{2n} + \frac{3n}{2}O_2 \rightarrow nCO_2 + nH_2O$ (for Alkene)

$x \qquad \frac{3n}{2}x \qquad nx$

$C_nH_{2n+2} + \frac{(3n+1)}{2}O_2 \rightarrow nCO_2 + (n+1)H_2O$ (for Alkane)

$(12-x) \qquad \frac{3n+1}{2}(12-x) \qquad (12-x)n$

Volume of $C_nH_{2n} = x$

Volume of $C_nH_{2n+2} = y$

Given $x + y = 12$

$nx + (12-x)n = 36$

$nx + 12n - nx = 36$

n=3

$$\frac{3n}{2}x + \frac{(3n+1)}{2}(12-x) = 57$$

$4.5x + 5(12-x) = 57$

$x = 6$

$y = 6$

48. Conceptual

49. $\mu_{cal} = e \times d$

$= 4.8 \times 10^{-10} \times 1.2 \times 10^{-8}$ cm

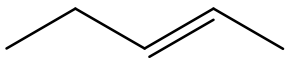
$= 5.76 \times 10^{-18}$ C.M

$= 5.76D$

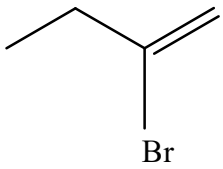
$$\text{Percentage of Ionic Character} = \frac{3.3D}{5.76D} \times 100 = 57.29\%$$

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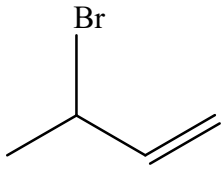
50.



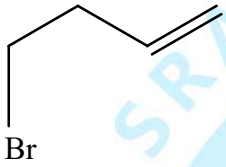
, 2



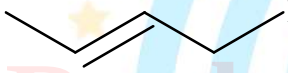
, 1



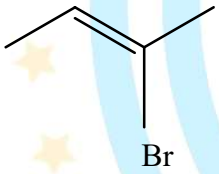
, 2



, 1



, 2



, 2

MATHS

51. $f(x)=\begin{cases}\dfrac{x}{1+x} & \text{if} \quad x \geq 0 \\ \dfrac{x}{1-x} & \text{if} \quad x<0\end{cases}$

For $x \geq 0: f'(x)=\dfrac{(1+x)(1)-x(1)}{(1+x)^2}=\dfrac{1}{(1+x)^2}>0$

For $x<0: f'(x)=\dfrac{(1-x)(1)-x(-1)}{(1-x)^2}=\dfrac{1}{(1-x)^2}>0$

Since the derivative is positive for all $x \in R$ (except a x = 0 where it is continuous and still increasing), the function is strictly increasing. A strictly increasing function is always one-one.

As $x \to \infty, f(x)=\dfrac{x}{1+x} \to 1$

As $x \to -\infty, f(x)=\dfrac{x}{1-x} \to -1$.

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Since the function is strictly increasing and its limits at infinity are 1 and – 1, the range of the functions is $(-1,1)$.

Because the range $(-1,1)$ is a subset of codomain R and not equal to it (e.g., there is no x such that f(x) = 5), the function is not onto.

52. $|A| = 3\cos\theta + 4\sin\theta$

Max value = $\sqrt{3^2+4^2} = \sqrt{25} = 5$

This occurs when $\tan\theta = \frac{4}{3}, \text{since } \frac{4}{3} > 0$, this value of θ lies within the interval $\left(0, \frac{\pi}{2}\right)$.

Behavior at end points:

$\theta \to 0^+, f(\theta) \to 3(1)+4(0) = 3$

$\theta \to \frac{\pi}{2}^-, f(\theta) \to 3(0)+4(1) = 4$

The range of the determinant is $[3,5]$

53. Total number of elements, n(A) = 999 – 100 + 1 = 900

Numbers divisible by 3 $(n(M_3)) = 300$

Numbers divisible by 5 $(n(M_5)) = 180$.

The number of elements divisible by either 3 or 5 is:

$n(M_3 \cup M_5) = n(M_3) + n(M_5) - n(M_{15})$

$n(M_3 \cup M_5) = 300 + 180 - 60 = 420$.

The number of elements in set B is the total number of in A minus those divisible by 3 or 5:

$n(B) = n(A) - n(M_3 \cup M_5)$

$n(B) = 900 - 420 = 480$

54. PA = PB

$(\alpha-3)^2 + (\beta-4)^2 = (\alpha-5)^2 + (\beta-(-2))^2$

$\Rightarrow \alpha = 3\beta + 1 \quad .....(1)$

Given Area = 10

$10 = \frac{1}{2}\left|3(-2-\beta) + 5(\beta-4) + \alpha(4-(-2))\right|$

$3\alpha + \beta = 23 \quad .....(2)$

$3\alpha + \beta = 3 \quad .....(3)$

$(1),(2)$ gives point P is $(7,2)$. This satisfies $\alpha\beta \neq 0$.

(1), (3) gives Point P is (1, 0). This does not satisfy $\alpha\beta \neq 0$.

So, $\alpha = 7$ and $\beta = 2$.

Point Q is $(4,3)$.

$d = \frac{|10|}{\sqrt{10}} = \sqrt{10}$

55. $\vec{a}.\hat{i} > 0, \vec{a}.\hat{j} > 0, \vec{a}.\hat{k} > 0$

56. $b_{3n} = b_1 r^{3n-1}$ and $b_{3n-2} = b_1 r^{3n-3}$

$b_1 r^2 \sum_{n=1}^{20} (r^3)^{n-1} = 4 b_1 r^0 \sum_{n=1}^{20} (r^3)^{n-1}$

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$$b_1 r^2 \sum_{n=1}^{20} \left(r^3\right)^{n-1} = 4b_1 \sum_{n=1}^{20} \left(r^3\right)^{n-1}$$

$$r^2 = 4$$

Since r is must be positive, we have r = 2.

$$S_{10} = 3\frac{2^{10}-1}{2-1} \Rightarrow S_{10} = 3069$$

57. Area of quadrilateral $OACB = P = \frac{1}{2}\left|\overline{OA}\times\overline{OB}\right| + \frac{1}{2}\left|\overline{OB}\times\overline{OC}\right|$

$$= \frac{1}{2}\left|\bar{a}\times\left(10\bar{a}+2\bar{b}\right)\right| + \frac{1}{2}\left|\bar{b}\times\left(10\bar{a}+2\bar{b}\right)\right| = 6\left|\bar{a}\times\bar{b}\right|$$

$$p = 6\left|\vec{a}\times\vec{b}\right|$$

$$q = \left|\overrightarrow{OA}\times\overrightarrow{OC}\right| = \left|\vec{a}\times\vec{b}\right|$$

$$\frac{p}{q} = \frac{6\left|\vec{a}\times\vec{b}\right|}{\left|\vec{a}\times\vec{b}\right|} = 6$$

58. $e^y dy = (1+x)\left(1+e^y\right)dx$

$$\int \frac{e^y}{1+e^y} dy = \int (1+x)dx$$

$$\ln\left(1+e^y\right) = x + \frac{x^2}{2} + C$$

$$y(1) = 0 \quad C = \ln(2) - \frac{3}{2}$$

$$x = -3;\ y = 0$$

59. The sum of 4 can be obtained by $\{(1,3),(2,2),(3,1)\}$. So, the probability that A throws a sum of 4 is:

$$P(A) = p = \frac{3}{36} = \frac{1}{12}$$

The probability that A does not throw a sum of 4 is $p' = 1 - \frac{1}{12} = \frac{11}{12}$.

The sum of 9 can be obtained by $\{(3,6),(4,5),(5,4),(6,3)\}$. So, the probability that B throws a sum of 9 is:

$$P(B) = q = \frac{4}{36} = \frac{1}{9}$$

The probability that B does not throw a sum of 9 is $q' = 1 - \frac{1}{9} = \frac{8}{9}$

$$P(A \text{ wins}) = q'p + q'p'(q'p) + q'p'\left(q'p'(q'p)\right) + ...$$

$$P(A \text{ wins}) = q'p\left[1 + q'p' + (q'p')^2 + ...\right]$$

$$= \frac{q'p}{1-q'p'} = \frac{2}{5}$$

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60. $f(x) = 5x^3 \cos x - x^2 \tan x + 2x^5 \tan x + 3x \sin x + 9x^4 \sin x$

Substitute the expression for f(x):

$$\frac{5x^3 \cos x - x^2 \tan x + 5x^4 \cos x - x^3 \tan x + 3x^2 \sin x + \text{higher order terms}}{x^3}$$

$$= 5\cos x - \frac{\tan x}{x} + 5x\cos x - \tan x + \frac{3\sin x}{x} + \text{higher order terms}$$

$$\lim_{x\to 0}\left(5\cos x - \frac{\tan x}{x} + 5x\cos x - \tan x + \frac{3\sin x}{x}\right)$$

$$= 5 - 1 + 0 - 0 + 3 = 7$$

61. $\frac{x^2}{4^2} + \frac{y^2}{1^2} = 1$

The rectangle must be bounded by the lines $x = \pm 4$ and $y = \pm 1$.

The vertices of this rectangle are $(\pm 4, \pm 1)$.

Let the equation of the outer ellipse be $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$

It passes through the point (16, 0).

$$\frac{16^2}{A^2} + \frac{0^2}{B^2} = 1 \Rightarrow A^2 = 16^2 = 256$$

Substitute one of the rectangle`s vertices, say (4, 1).

$$\frac{1}{B^2} = 1 - \frac{1}{16} = \frac{15}{16}$$

$$x^2 + 240y^2 = 16^2$$

62. $t = \log_e x,\ x = e^t,\ dx = e^t dt$

$$f(x) = \int e^t \left(\frac{1}{t} - \frac{2}{t^3}\right) dt$$

$$f(x) = \int e^t \left(\frac{1}{t} - \frac{1}{t^2}\right) dt + \int e^t \left(\frac{1}{t^2} - \frac{2}{t^3}\right) dt$$

$$f(x) = \frac{e^t}{t} + \frac{e^t}{t^2} + C$$

$$f(e) = 2e \Rightarrow C = 0$$

$$f(e^2) = \frac{2e^2 + e^2}{4} = \frac{3e^2}{4}$$

63. The data set has 10 values: 4,10,6,4,13,11,19, m,8,7. We are given $m \geq 11$.

$$M = \frac{8+10}{2} = \frac{18}{2} = 9$$

The formula for mean deviation about the median is $\frac{\sum |x_i - M|}{N}$.

$$\frac{|-5| + |-5| + |-3| + |-2| + |-1| + |1| + |2| + |4| + |10| + |m-9|}{10} = 4.$$

We are given that $m \geq 11$. Therefore, m = 16 is the correct value.

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64. Let the position vector of point A be $\vec{a} = p\left(\hat{i}+\hat{j}+\hat{k}\right)$.

The first line L_1 passes through P(0, 0, 1) and is parallel to $\vec{d}_1 = \hat{i}+\hat{j}$.

The second line L_2 passes through Q(0, 0, -1) and is parallel to $\vec{d}_2 = \hat{i}-\hat{j}$.

Point B lies on L_1, so its position vector is $\vec{b} = \hat{k}+\lambda\left(\hat{i}+\hat{j}\right) = \lambda\hat{i}+\lambda\hat{j}+\hat{k}$ for some scalar λ.

$\overrightarrow{AB}\,.\,\vec{d}_1 = 0$:

$$(\lambda-p)(1)+(\lambda-p)(1)+(1-p)(0)=0$$
$$2(\lambda-p)=0 \Rightarrow \lambda = p$$

So the position vector of B is $\vec{b} = p\hat{i}+p\hat{j}+\hat{k}$.

Point C lies on L_2, so its position vector is $\vec{c} = -\hat{k}+\mu\left(\hat{i}-\hat{j}\right) = \mu\hat{i}-\mu\hat{j}-\hat{k}$ for some scalar μ.

Since AC is perpendicular to L_2, AC. $d_2 = 0$

$$(\mu-p)(1)+(-\mu-p)(-1)+(-1-p)(0)=0$$
$$\mu-p+\mu+p=0$$
$$2\mu=0 \Rightarrow \mu=0$$

So the position vector of C is $\vec{c} = -\hat{k}$.

$\overrightarrow{AB} = (p-p)\hat{i}+(p-p)\hat{j}+(1-p)\hat{k} = (1-p)\hat{k}$

$\overrightarrow{AC} = (0-p)\hat{i}+(0-p)\hat{j}+(-1-p)\hat{k} = -p\hat{i}-p\hat{j}-(1+p)\hat{k}$

Since $\overrightarrow{AB}\,.\,\overrightarrow{AC} = 0 : p = \pm 1$

65. $d = r_1 + r_2$ or $d = |r_1 - r_2|$

66. $(1+10)^{2016} = \binom{2016}{0}10^{2016}+\binom{2016}{1}10^{2015}+.....+\binom{2016}{2014}10^2+\binom{2016}{2015}10^1+\binom{2016}{2016}10^0$

The sum of last three terms is:

$2031120\times100+2016\times10+1\times1 = \quad 203112000+20160+1$

$=.....12000+160+1=161$

The digit at the tens place is x = 6.

The digit at the hundreds place is y = 1.

67. $\arg(z)=\frac{\pi}{3}$ is $y=\left(\tan\frac{\pi}{3}\right)x=\sqrt{3}\,x$ for $x\geq 0$

$\left|z-2\sqrt{3}\,i\right|=2$ is $x^2+\left(y-2\sqrt{3}\right)^2=2^2=4$

Solving

$$x^2+\left(\sqrt{3}x-2\sqrt{3}\right)^2=4$$

The intersection points are $A=\left(1,\sqrt{3}\right)$ and $B=\left(2,2\sqrt{3}\right)$.

The x-coordinates of the intersection points are $x_1=1$ and $x_2=2$.

The intersection points are $A = (1, \sqrt{3})$ and $B = (2, 2\sqrt{3})$.

AB = 2

68. $f(x)=x^3+e^x+5$

$$f'(x)=\frac{d}{dx}\left(x^3+e^x+5\right)=3x^2+e^x$$

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$f'(x) > 0$ for all $x \in R$

As $x \to -\infty : x^3$ becomes a very large negative number, e^x approaches 0, and 5 remains constant. Therefore, $f(x) \to -\infty$.

As $x \to \infty$: Both x^3 and e^x approach infinity. Therefore, $f(x) \to \infty$.

Therefore there is exactly one root.

69. Let the three quadratic expressions be:

$f_1(x) = ax^2 + bx + c$

$f_2(x) = ax^2 - 2bx - c$

$f(x) = ax^2 + 4bx + 2c$

Given that 2α is a root of $f_1(x)$, we have:

$a(2\alpha)^2 + b(2\alpha) + c = 0 \Rightarrow 4\alpha a^2 + 2b\alpha + c = 0$

$2c = -8a\alpha^2 - 4b\alpha$

Given that β is a root of $f_2(x)$, we have:

$a\beta^2 - 2b\beta - c = 0 \Rightarrow c = a\beta^2 - 2b\beta$

$f(\alpha) = a\alpha^2 + 4b\alpha + \left(-8a\alpha^2 - 4b\alpha\right)$

$f(\alpha) = -7a\alpha^2$

Since $a > 0$ and $\alpha^2 > 0$, then $f(\alpha) < 0$

$f(\beta) = a\beta^2 + 4b\beta + \left(2a\beta^2 - 4b\beta\right)$

$f(\beta) = 3a\beta^2$

Since $a > 0$ and $\beta^2 > 0$, then $f(\beta) > 0$

$\therefore \beta < \gamma < \alpha$

70. The endpoints of a focal chord are $P(t^2, 2t)$ and $Q(1/t^2, -2/t)$.

The center of the circle with diameter PQ is the midpoint of PQ.

The coordinates of the midpoint M are:

$$M = \left(\frac{t^2 + \frac{1}{t^2}}{2}, \frac{2t + \left(-\frac{2}{t}\right)}{2}\right) = \left(\frac{t^2 + \frac{1}{t^2}}{2}, t - \frac{1}{4}\right)$$

The center M lies on the line $\sqrt{5}y + 4 = 0$.

$$\sqrt{5}\left(t - \frac{1}{t}\right) + 4 = 0$$

The length of the focal chord PQ is given by the formula $L = a\left(t + \frac{1}{t}\right)^2$.

$$L = a\left(\left(t - \frac{1}{t}\right)^2 + 4\right) = 1.\left(\left(-\frac{4}{\sqrt{5}}\right)^2 + 4\right) = \frac{36}{5}$$

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71. $y = 4^{\cos^2\alpha}$

$y + \frac{4}{y} = 4 \Rightarrow y = 2$

$2\cos^2\alpha = 1 \Rightarrow \cos^2\alpha = \frac{1}{2},\ \cos\alpha \pm \frac{1}{\sqrt{2}}$

Thus, $S = \left\{\frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\right\}$

$\sum_{\alpha\in S} \cot\alpha = (-1) + 1 + (-1) = -1$

72. $T_n = S_n - S_{n-1}$

$T_n = 4n(n+1)(n+2)$

$\sum_{r=1}^{10} \frac{1}{T_r} = \sum_{r=1}^{10} \frac{1}{8}\left(\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)}\right)$

$= \frac{1}{8}\left[\frac{1}{1\times 2} - \frac{1}{11\times 12}\right] = \frac{65}{1056}$

73. $21_{P_2}.4.9 = 15120$

74. The total area (A) is the integral of the curve $y = 1 + 4x - x^2$ from $x = 0$ to $x = 3/2$.

$A = \int_0^{3/2} (1 + 4x - x^2)dx = \frac{39}{8}$

$\text{Area}_{\text{bisected}} = \int_0^{3/2} mx\ dx = \frac{9m}{8}$

$\frac{9m}{8} = \frac{39}{16} \Rightarrow m = \frac{13}{6}$

75. $I = \int_{-3/2}^{3/2} f(x)dx = \int_{-3/2}^{3/2} [x^2]dx + \int_{-3/2}^{3/2} g(x)\,dx$

We evaluate this integral using the piecewise definition of g(x) provided:

1. For $-\frac{3}{2} \le x < -1$: $\int_{-3/2}^{-1} (-2x - 3)dx = -0.25$

2. For $-1 \le x \le \frac{3}{2}$: $\int_{-1}^{1} x\,dx = 0$

3. For $1 \le x \le \frac{3}{2}$: $\int_{1}^{3/2} (3 - 2x)^{dx} = 0.25$

$\int_{-3/2}^{3/2} g(x)\,dx = -0.25 + 0 + 0.25 = 0$

$\int_{-3/2}^{3/2} [x^2]dx = 2\int_{0}^{3/2} [x^2]dx = 4 - 2\sqrt{2}$

$I = (4 - 2\sqrt{2})$

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